- GLA University recently concluded its annual campus placement drive, aiming to help graduating students secure job opportunities. During the process, the university collected detailed information about each student to analyze the success of the drive and identify areas for improvement. They recorded each student's department, such as Computer Science, Mechanical, Electrical, or Civil, and their placement status, indicating whether they were placed or not. For students who received job offers, the university noted the rank of the company that made the offer, based on the university's internal reputation index. To understand placement outcomes further, the university tracked the annual salary package offered (in lakhs per annum) and the total time each student spent in interviews (in minutes). Additionally, students were asked to provide a feedback rating on the placement process, ranging from 1 (Very Dissatisfied) to 5 (Very Satisfied). The university also calculated the percentage of students placed from each department and the average number of companies visited by students from different departments. Identify the variables and discuss about the scale of measurement for each variable.
- David's school conducted a survey to understand students' preferences and academic performance. The survey included questions about each student's favorite subject (Math, Science, History, or English) and their grade level (e.g., 9th, 10th, 11th, or 12th). The school also recorded the marks obtained in the last math exam (out of 100) and the time spent studying daily (in hours). As part of the survey, students were asked to rank their teachers' helpfulness (e.g., Teacher A > Teacher B > Teacher C). Additionally, the school tracked the number of extracurricular activities each student participated in and their attendance percentage for the semester. List the variables and determine the scale of measurement for each variable.
- Samantha participated in a survey conducted by a retail store to understand shopping habits. The store recorded Samantha's preferred mode of shopping (Online or In-store) and her age in years. During the survey, she was asked to rate her satisfaction with the store on a scale from 1 (Very Unsatisfied) to 5 (Very Satisfied). The researchers tracked her monthly spending in dollars and the number of items purchased per visit over the past three months. Samantha was also asked to list her top three preferred product categories (e.g., Electronics > Clothing > Groceries). List the variables and determine the scale of measurement for each variable.
- Maria is participating in a health and wellness study conducted by a research institute. As part of the study, the researchers recorded several details about her to analyze her health and activity levels. They noted her blood group (A, B, AB, or O) and her age in years. Maria's weight in kilograms and height in centimeters were measured during her initial visit. She was asked to rate the program on a scale from 1 (Very Poor) to 5 (Excellent). Over the course of a week, Maria tracked the number of steps she walked daily and the time in minutes she spent exercising each day. The researchers also recorded the department she visited for her check-up (e.g., General Medicine, Cardiology, or

Nutrition). Identify the variables and discuss about the scale of measurement for each variable.

- David is participating in a fitness challenge and is monitoring his daily steps for a month. Over the 30 days, he recorded the following number of steps per day:
  5200, 6800, 7100, 4900, 5300, 8000, 9200, 6500, 5800, 7400, 5600, 8300, 6200, 7700, 9000, 6400, 5500, 8200, 7300, 9100, 5900, 6300, 4800, 7200, 8100, 6600, 7500, 9300, 6700, 8400.
  Construct a grouped frequency distribution for David's daily steps using appropriate class intervals (e.g., intervals of 1,000 steps). Calculate the frequency for each class interval.
- A teacher wants to analyze the performance of her class of 50 students on a recent math exam. The scores (out of 100) obtained by the students are as follows:
  48, 65, 78, 82, 53, 91, 70, 69, 45, 67, 76, 88, 59, 72, 81, 85, 50, 62, 77, 68, 73, 79, 84, 92, 60, 58, 47, 64, 83, 66, 90, 52, 71, 75, 80, 54, 89, 63, 74, 87, 61, 55, 46, 68, 49, 56, 86, 93, 57, 66.
  Organize the data into a grouped frequency distribution table using class intervals of 10 marks. Calculate the cumulative frequency for each class interval.

- A financial advisor is conducting a survey to understand the monthly expenses of 40 households. The expenses (in dollars) are as follows:
  800, 950, 1200, 1100, 850, 1050, 1400, 1250, 900, 1150, 1300, 1600, 1450, 1000, 850, 1050, 1200, 1100, 950, 1300, 1500, 1400, 1150, 1050, 1250, 1000, 1200, 850, 1100, 1300, 1400, 1150, 1000, 1600, 1300, 1050, 950, 1250, 1450.
  Create a grouped frequency distribution table for the monthly expenses using class intervals of 100 dollars. Identify the range of expenses that is most common among households.
- A bank is analyzing customer waiting times to improve its service. The waiting times (in minutes) of 40 customers were recorded as follows:
  5, 10, 12, 15, 20, 8, 9, 18, 25, 30, 7, 22, 14, 11, 16, 13, 17, 19, 21, 28, 6, 24, 29, 26, 31, 33, 27, 20, 12, 23, 15, 22, 10, 8, 13, 11, 14, 19, 17, 21.
  The bank decides to group the waiting times into intervals of 5 minutes and calculate the frequency for each interval. To better understand the data, the bank plans to create both a histogram and an ogive.
- GLA University conducted a campus placement drive for final-year students. The Career Services team recorded the annual salary packages (in lakhs per annum) offered to 50 students. The salary packages are as follows:
  3, 5, 6, 8, 7, 9, 10, 5, 12, 6, 8, 11, 4, 9, 13, 14, 6, 7, 8, 10, 3, 12, 5, 7, 9, 6, 11, 10, 8, 5, 9, 6, 14, 12, 7, 8, 13, 5, 9, 10, 4, 6, 7, 11, 10, 3, 12, 6, 8, 9.
  Organize the salary data into a grouped frequency distribution table with intervals of 2 LPA. Draw a histogram to represent the distribution of salary packages. Construct a "less than" ogive for the cumulative frequency.

- A class of 15 students participated in a study to determine the relationship between the number of hours they studied and their performance in a recent exam. The teacher recorded the number of hours each student studied per week and their percentage of marks scored in the exam. The data collected is as follows:
  Study Hours (per week): 2, 4, 3, 5, 6, 7, 3, 4, 8, 5, 6, 7, 4, 3, 6
  Percentage of Marks: 55, 65, 60, 70, 75, 80, 58, 66, 85, 72, 78, 82, 68, 61, 76
  Plot a scatter plot with the number of study hours on the x-axis and the percentage of marks on the y-axis.
- A sports coach is studying the relationship between the age and height of 15 young athletes to understand growth patterns. The coach recorded the ages (in years) and heights (in centimeters) of the athletes. The data is as follows:
  Age (Years): 10, 11, 12, 13, 14, 15, 16, 17, 18, 12, 14, 16, 10, 13, 15
  Height (cm): 140, 145, 150, 155, 160, 162, 165, 170, 175, 152, 158, 168, 142, 157, 163
  Plot a scatter plot with age on the x-axis and height on the y-axis.
- A financial advisor is analyzing the spending habits of 15 individuals by comparing their monthly income and monthly expenses (both in thousands). The data collected is as follows:
  Monthly Income (in ₹ '000): 40, 45, 50, 55, 60, 65, 70, 75, 80, 50, 60, 70, 55, 45, 65
  Monthly Expenses (in ₹ '000): 30, 35, 40, 45, 50, 55, 60, 65, 70, 38, 48, 68, 42, 33, 58
  Plot a scatter plot with monthly income on the x-axis and monthly expenses on the y-axis.
- A school principal conducted a survey to analyze the academic performance of students across four different classes (10th, 11th, 12th Science, and 12th Commerce). The principal recorded the average marks scored by students in their respective classes in the previous semester. The data collected includes the number of students in each class and their average marks: 10th: 45 students, average marks 72, 11th: 50 students, average marks 65, 12th Science: 40 students, average marks 78, 12th Commerce: 35 students, average marks 70. Identify the type of data collected (cross-section or time series). Explain why this data is considered cross-sectional.
- A retail store is tracking its monthly sales for one year to analyze trends and make future predictions. The sales (in thousands of dollars) for each month are as follows: January: 40, February: 42, March: 38, April: 45, May: 48, June: 50, July: 52, August: 49, September: 46, October: 53, November: 55, December: 60.
  Identify whether the data is cross-sectional or time series. Plot the data on a time series graph to visualize the monthly sales trend.
- An agricultural department is studying the production of wheat across four regions over the past five years. The data includes the **annual wheat production** (in tons) for each region:
  Region A: 2018: 150, 2019: 160, 2020: 155, 2021: 165, 2022: 170
  Region B: 2018: 120, 2019: 125, 2020: 130, 2021: 140, 2022: 150
  Region C: 2018: 110, 2019: 115, 2020: 120, 2021: 125, 2022: 130
  Region D: 2018: 180, 2019: 190, 2020: 185, 2021: 195, 2022: 200
  Identify which part of the data is cross-sectional and which part is time series. Visualize the **time series data** for each region using line graphs.